Mar 11, 2024

textbook : 7.8 Improper integrals < infinite integrals (ex 1, 6)
discontinuous integrals (ex 2, 3a, 3b, 6)

8.1 Arc length (ex 4a)

8.2 Surface area of surface of rotation (ex 4b, 5)

ex 1: $\displaystyle\int_{-\infty}^{-5} \frac{x}{(x^2-4)^{5/2}}\,dx = \lim_{t\to-\infty} \int_t^{-5} \frac{x}{(x^2-4)^{5/2}}\,dx$

$\left[\begin{array}{l|l} u = x^2-4 & (-5)^2-4 = 21 \\ du = 2x\,dx & t^2-4 \end{array}\right.$

$$= \lim_{t\to-\infty} \int_{t^2-4}^{21} \frac{du}{2u^{5/2}}$$

$$= \lim_{t\to-\infty} \left[\frac{1}{\cancel{2}}\left(-\frac{\cancel{2}}{3}\right)\frac{1}{u^{3/2}}\right]_{t^2-4}^{21}$$

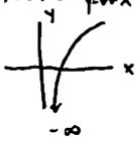
$$= \lim_{t\to\infty} \left[-\frac{1}{3(21)^{3/2}} + \frac{1}{3(t^2-4)^{3/2}}\right]$$

$$= \boxed{-\frac{1}{3(21)^{3/2}}}$$

ex 2: $\displaystyle\int_0^3 x\ln x\,dx = \lim_{t\to0^+} \int_t^3 x\ln x\,dx$

$\left[\begin{array}{ll} u = \ln x & v = \frac{x^2}{2} \\ du = \frac{1}{x}dx & dv = x\,dx \end{array}\right]$

Note: $y = \ln x$



$$= \lim_{t\to0^+} \left[\frac{x^2}{2}\ln x\Big|_t^3 - \int_t^3 \frac{x^{\cancel{2}}}{2}\cdot\frac{1}{\cancel{x}}\,dx\right]$$

$$= \lim_{t\to0^+} \left[\frac{x^2}{2}\ln x\Big|_t^3 - \frac{x^2}{4}\Big|_t^3\right]$$

$$= \lim_{t\to0^+} \left[\frac{9}{2}\ln 3 - \frac{t^2}{2}\ln t - \frac{9}{4} + \frac{t^2}{4}\right]$$

Note $\displaystyle\lim_{t\to0^+}\frac{t^2}{4} = 0$.

$$= \frac{9}{2}\ln 3 - \frac{9}{4} - \frac{1}{2}\underbrace{\lim_{t\to0^+}(t^2\ln t)}$$

$$\lim_{t\to0^+}(t^2\ln t) = \lim_{t\to0^+}\left(\frac{\ln t}{1/t^2}\right)$$

use L'Hopital's Rule

$$= \lim_{t\to0^+}\left(\frac{1/t}{-2/t^3}\right)$$

$$= \lim_{t\to0^+}\left(\frac{1}{t}\cdot\frac{t^3}{-2}\right)$$

$$= \lim_{t\to0^+}\left(\frac{t^2}{-2}\right)$$

$$= 0$$

So, $\displaystyle\int_0^3 x\ln x\,dx = \boxed{\frac{9}{2}\ln 3 - \frac{9}{4}}$

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ex 3a: $\displaystyle\int_{-1}^{2} \frac{1}{\sqrt{2-x}}\,dx = \lim_{t\to2^-}\int_{-1}^{t}\frac{1}{\sqrt{2-x}}$

$$= \lim_{t\to2^-}\left[-2\sqrt{2-x}\right]_{-1}^{t}$$

$$= \lim_{t\to2^-}\left[-2\sqrt{2-t} + 2\sqrt{3}\right]$$

$$= \boxed{2\sqrt{3}}$$

or, using a u-sub

$\left[\begin{array}{l|l} u = 2-x & 2-t \\ du = -dx & 2-(-1) = 3 \end{array}\right.$

$$\lim_{t\to2^-}\int_3^{2-t}\left(-\frac{1}{\sqrt{u}}\,du\right)$$

$$= \lim_{t\to2^-}\left[-2\sqrt{u}\right]_3^{2-t}$$

$$= \lim_{t\to2^-}\left[-2\sqrt{2-t} + 2\sqrt{3}\right]$$

$$= \boxed{2\sqrt{3}}$$

ex 3b: Note that improper integrals may diverge:

$$\int_{-1}^{2}\frac{1}{(2-x)^{3/2}}\,dx$$

$$= \lim_{t\to2^-}\int_{-1}^{t}\frac{1}{(2-x)^{3/2}}\,dx$$

$$= \lim_{t\to2^-}\left(\frac{2}{\sqrt{2-x}}\right)\Big|_{-1}^{t}$$

$$= \lim_{t\to2^-}\left(\underbrace{\frac{2}{\sqrt{2-t}}}_{\to\infty \text{ as } t\to2^-} - \frac{2}{\sqrt{3}}\right)$$

$\boxed{\text{Diverges}}$

ex 4a: Use arc length formula: $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

Find length of curve $y = \frac{x^3}{4} + \frac{1}{3x}$ for $1 \le x \le 4$.

$$\frac{dy}{dx} = \frac{3}{4}x^2 - \frac{1}{3x^2}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{9}{16}x^4 - 2\left(\frac{1}{4}\right) + \frac{1}{9x^4}$$

$$= \frac{9}{16}x^4 - \frac{1}{2} + \frac{1}{9x^4}$$

$$L = \int_1^4 \sqrt{1 + \frac{9}{16}x^4 - \frac{1}{2} + \frac{1}{9x^4}}\, dx$$

$$= \int_1^4 \sqrt{\frac{9}{16}x^4 + \frac{1}{2} + \frac{1}{9x^4}}\, dx$$

$$= \int_1^4 \sqrt{\left(\frac{3}{4}x^2 + \frac{1}{3x^2}\right)^2}\, dx$$

$$= \int_1^4 \left(\frac{3}{4}x^2 + \frac{1}{3x^2}\right) dx$$

$$= \left[\frac{x^3}{4} - \frac{1}{3x}\right]_1^4$$

$$= \frac{4^3}{4} - \frac{1}{12} - \left(\frac{1}{4} - \frac{1}{3}\right)$$

$$= \boxed{16}$$

ex 4b: Rotate the curve from ex 4a about the x-axis:

$$S = \int_1^4 2\pi\left(\frac{x^3}{4} + \frac{1}{3x}\right)\sqrt{1 + \frac{9}{16}x^4 - \frac{1}{2} + \frac{1}{9x^4}}\, dx$$

$$= 2\pi \int_1^4 \left(\frac{x^3}{4} + \frac{1}{3x}\right)\left(\frac{3}{4}x^2 + \frac{1}{3x^2}\right) dx$$

$$= 2\pi \int_1^4 \left(\frac{3}{16}x^5 + \frac{x}{12} + \frac{x}{4} + \frac{1}{9x^3}\right) dx$$

$$= 2\pi \int_1^4 \left(\frac{3}{16}x^5 + \frac{x}{3} + \frac{1}{9x^3}\right) dx$$

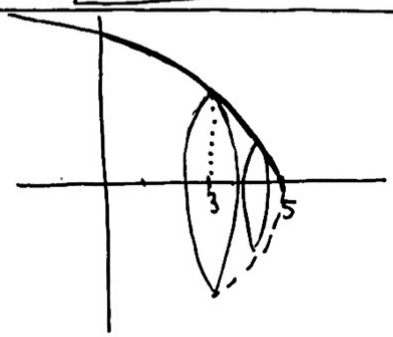
$$= 2\pi \left[\frac{\cancel{3}}{16} \cdot \frac{x^6}{\cancel{6}_2} + \frac{x^2}{6} + \frac{1}{9}\left(\frac{1}{-2x^2}\right)\right]_1^4$$

$$= 2\pi \left[\frac{1}{32}(4^6 - 1) + \frac{15}{6} - \frac{1}{18}\left(\frac{1}{16} - 1\right)\right]$$

$$= 2\pi \left[\frac{1}{32}(4^6 - 1) + \frac{15}{6} - \frac{1}{\cancel{18}_6}\left(-\frac{\cancel{15}^5}{16}\right)\right]$$

$$= 2\pi \left[\frac{3 \cdot 4^6 - 3 + 15 \cdot 16 + 5}{96}\right]$$

$$= \boxed{\frac{6265\pi}{24}}$$



ex 5: $S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

Find surface area of the surface made by rotating $y = \sqrt{5-x}$, $3 \le x \le 5$, about x-axis

$$\frac{dy}{dx} = \frac{1}{2\sqrt{5-x}}(-1)$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{1}{4(5-x)}$$

$$S = \int_3^5 2\pi\sqrt{5-x}\sqrt{1 + \frac{1}{4(5-x)}}\, dx$$

$$= 2\pi \int_3^5 \sqrt{(5-x)\left(1 + \frac{1}{4(5-x)}\right)}\, dx$$

$$= 2\pi \int_3^5 \sqrt{5 - x + \frac{1}{4}}\, dx$$

$$= 2\pi \int_3^5 \sqrt{\frac{21}{4} - x}\, dx$$

$$\left[\begin{array}{l|l} u = \frac{21}{4} - x & \frac{21}{4} - 5 = \frac{1}{4} \\ du = -dx & \frac{21}{4} - 3 = \frac{9}{4} \end{array}\right.$$

$$= -2\pi \int_{9/4}^{1/4} \sqrt{u}\, du$$

$$= 2\pi \int_{1/4}^{9/4} \sqrt{u}\, du$$

$$= 2\pi \cdot \frac{2}{3}u^{3/2}\Big|_{1/4}^{9/4}$$

$$= \frac{4\pi}{3}\left[\left(\frac{9}{4}\right)^{3/2} - \left(\frac{1}{4}\right)^{3/2}\right]$$

$$= \frac{4\pi}{3}\left[\frac{27}{8} - \frac{1}{8}\right]$$

$$= \boxed{\frac{13\pi}{3}}$$

ex 6: $\int_{-\infty}^{5} \frac{x}{(x^2-4)^{5/2}}\,dx$ is an infinite integral with 2 discontinuities (at $x=-2$, $x=+2$).

So, we need to split up the interval $(-\infty, 5]$ appropriately so that we can integrate:

$$\int_{-\infty}^{-5} \frac{x}{(x^2-4)^{5/2}}\,dx + \int_{-5}^{-2} \frac{x}{(x^2-4)^{5/2}}\,dx + \int_{-2}^{0} \frac{x}{(x^2-4)^{5/2}}\,dx + \int_{0}^{2} \frac{x}{(x^2-4)^{5/2}}\,dx + \int_{2}^{5} \frac{x}{(x^2-4)^{5/2}}\,dx$$

Note: We want only 1 "problem" per integral (i.e., infinity or discontinuity). I chose to split up the integral this way for convenience, but you don't have to use -5 and 0 as in the first 2 and 2nd 2 integrals.

We already integrated the 1st integral in ex 1, so we know it converges and is equal to $-\frac{1}{3(21)^{3/2}}$. So, let's look at the 2nd:

$$\int_{-5}^{-2} \frac{x}{(x^2-4)^{5/2}}\,dx = \lim_{t\to -2^-} \int_{-5}^{t} \frac{x}{(x^2-4)^{5/2}}\,dx = \lim_{t\to -2^-} \int_{21}^{t^2-4} u^{-5/2}\,\frac{du}{2}$$ (same u-sub as in ex 1)

$$= \lim_{t\to -2^-} \left[-\frac{1}{3}\cdot\frac{1}{u^{3/2}}\right]_{21}^{t^2-4} = \lim_{t\to -2^-} \left[-\frac{1}{3}\cdot\frac{1}{(t^2-4)^{3/2}} + \frac{1}{3(21)^{3/2}}\right]$$ DIVERGES

as $t\to -2^-$, $t^2-4 \to 0$ so this fraction approaches ∞

Since one of the terms in our sum of integrals diverges, the whole sum is divergent.

So, $\int_{-\infty}^{5} \frac{x}{(x^2-4)^{5/2}}\,dx$ $\boxed{\text{diverges}}$.